TakeSure — Confidence before taking something

Scan both. Check the pair. Ask the right questions.

An Ayurveda × Allopathy herb–drug interaction checker for Indian patients, doctors and the Ministry of Ayush

HACK@BVP 8.0 · Abstract Round
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DOMAIN: Health & Well-Being (AI4Impact secondary) · SDG 3 · SDG 9 · SDG 10

One patient's story — the gap today, and what TakeSure would do about it

1. **Mother** takes Thyronorm for thyroid and starts ashwagandha for stress.
2. **Nobody asks, nobody tells.** Her doctor never hears of it; the label says nothing.
3. **She photographs both.** TakeSure reads the names, confirms them and checks the pair.
4. **AMBER.** A cited source, timing advice, three questions. One tap sends the PDF to her doctor.

1 The problem

- **Two medical systems, zero cross-checking.** A thyroid patient on Thyronorm adds ashwagandha; a transplant patient on tacrolimus drinks giloy kadha; a cardiac patient on warfarin takes guggul.
- **Each pair can cause harm:** ashwagandha can raise thyroid hormone, giloy can oppose immunosuppression, guggul can increase bleeding.
- **Nobody asks, nobody tells.** The doctor rarely asks about herbs, the patient rarely mentions them, the chemist has no tool, the label says nothing.
- **Existing tools do not cover it.** Medscape, Drugs.com and 1mg check only allopathic × allopathic pairs; Western herb databases are English-only, paid and do not know Indian brands like Himalaya Ashvagandha or Dabur Giloy Ghanvati.

37%	of hospital patients use allopathic and herbal medicines together	**65%**	of herbal users do not know the dose, side effects or interactions
19%	of Indian patients ever disclose herbal use to their doctor	**1,473**	ASU&H adverse-event reports received nationally since 2018

2 Proposed solution

- **Scan both.** The patient photographs the allopathic prescription and the Ayurvedic product label (or types the names); every name is confirmed before checking.
- **Match Indian names.** Brand → generic (Thyronorm → levothyroxine) and product → herbs (Chyawanprash → amla, ashwagandha…) with Sanskrit / Hindi / Latin synonyms.
- **Decide from data.** A rule engine checks every herb–drug pair against a curated, cited knowledge base: GREEN / AMBER / RED / NOT IN DATABASE, with timing advice.
- **Explain simply.** AI rewrites only the matched evidence into plain Hindi or English, with sources and three questions to ask the doctor.
- **Report in one tap.** A one-page PDF for the doctor and a pre-filled suspected-ADR report in the Ministry of Ayush format for the Ayush Suraksha portal.

Our rule: The verdict comes from published data; AI only explains it.

3 Approach — project workflow

Prescription photo allopathic medicines + **Product photo** Ayurvedic label / brand → **1 · OCR** read both images; user confirms names → **2 · Match** brand → generic; product → herbs; synonyms → **3 · Knowledge base** cited herb–drug table; grade A/B/C → **Rule engine** decides severity; AI does not → GREEN / AMBER / RED / NOT IN DB → **4 · AI explains** simple Hindi/English + 3 doctor questions → **Doctor PDF** verdict + sources; **ADR report** Ministry of Ayush format

Verdict rules: Worst row wins; an unknown pair is never shown as green; grade-C evidence caps at AMBER; the answer is identical with the AI switched off.

Feature	Medscape · Drugs.com · 1mg	MSKCC · Natural Medicines	TakeSure
Ayurveda × allopathy pair check	✗	~ single herbs	✓ every pair
Indian brands & Hindi	~	✗	✓ brands + synonyms
Source on every verdict	✓	✓	✓ + evidence grade
ADR report to Ministry of Ayush	✗	✗	✓ one tap

Our assessment from each tool's public documentation.

Stack. Next.js + TypeScript PWA (Hindi/English UI) · FastAPI + Pydantic REST API · PostgreSQL in production, SQLite offline · Tesseract OCR · RapidFuzz matching (≥85 accept, 70–85 ask the user) · Groq LLM, JSON-schema output, temperature 0 · ReportLab PDFs. No model training; runs fully offline on a laptop.

Knowledge base. 40–60 herb × drug pairs curated from PubMed, NIH LiverTox, MSKCC, WHO monographs and the Ayurvedic Pharmacopoeia, graded on the Stockley's model. Decision support only, never diagnosis. Sample rows:

Herb × drug	Verdict	What happens	Source
Ashwagandha × Levothyroxine	AMBER · grade B	May raise thyroid hormone	Sharma 2018, PMID 28829155
Guggul × Warfarin	RED · grade B	May increase bleeding risk	MSKCC About Herbs
Giloy × Tacrolimus	RED · grade C	May oppose immuno-suppression	Nagral 2021, LiverTox

4 Expected impact

- **Patients & caregivers** A sourced answer before the first dose, in their own language, instead of silence or a guess. Free for citizens.
- **Doctors, vaidyas & chemists** Finally learn what patients actually take; a QR on the chemist's bill or an API inside hospital and telemedicine systems makes the check routine.
- **Ministry of Ayush** Structured, reproducible ADR reports and real co-use data from the catchment of its 99 peripheral pharmacovigilance centres.
- **Novelty** Nothing public today checks an Ayurvedic product against an allopathic prescription. TakeSure builds the missing bridge table: Indian brands × Ayurvedic herb synonyms × graded evidence, open and auditable. Honest by design: grades, timing gaps instead of alarms, "not in database" instead of guesses.